LAB ASSIGNMENT – 9.5

NAME – Anushka Boora

Ht No.: 2303A510E6

BATCH – 29

Problem 1: String Utilities Function

Consider the following Python function:

```
def reverse_string(text):
return text[::-1]
```

Task:

1. Write documentation in:

o (a) Docstring

o (b) Inline comments

o (c) Google-style documentation

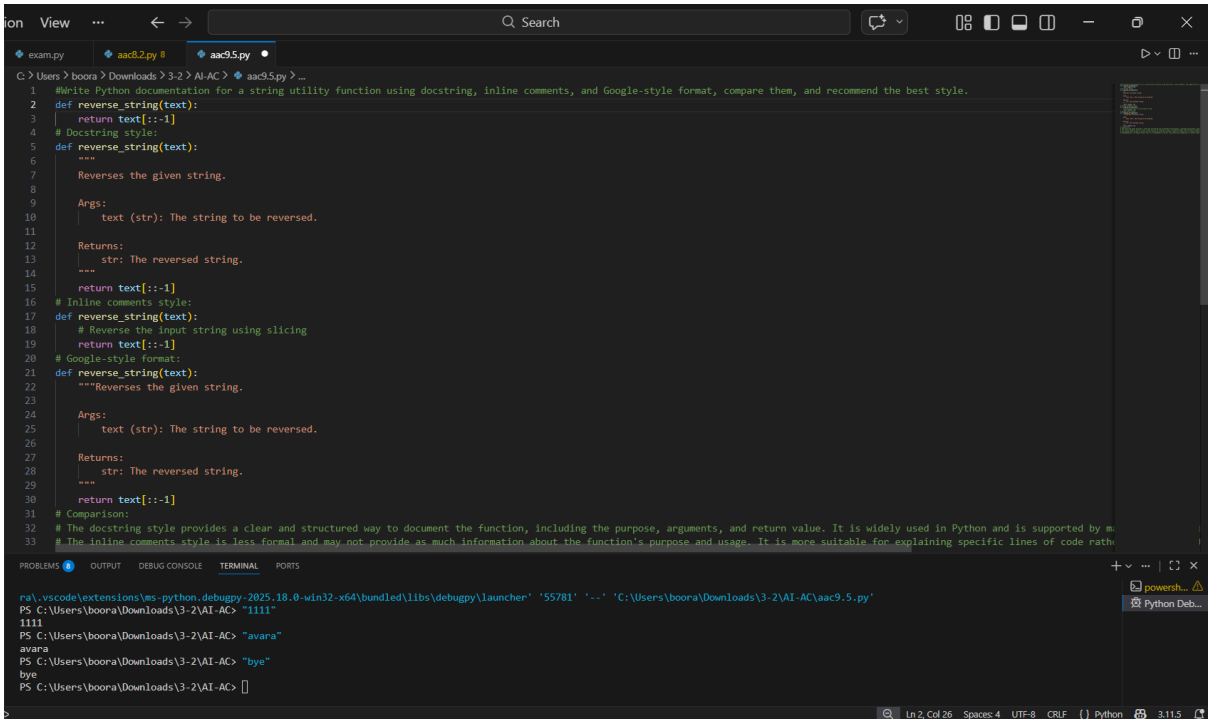
2. Compare the three documentation styles.

3. Recommend the most suitable style for a utility-based string

library.

Prompt:

#Write Python documentation for a string utility function using docstring, inline comments, and Google-style format, compare them, and recommend the best style.



Problem 2: Password Strength Checker

Consider the function:

def check_strength(password):

return len(password) >= 8

Task:

1. Document the function using docstring, inline comments, and

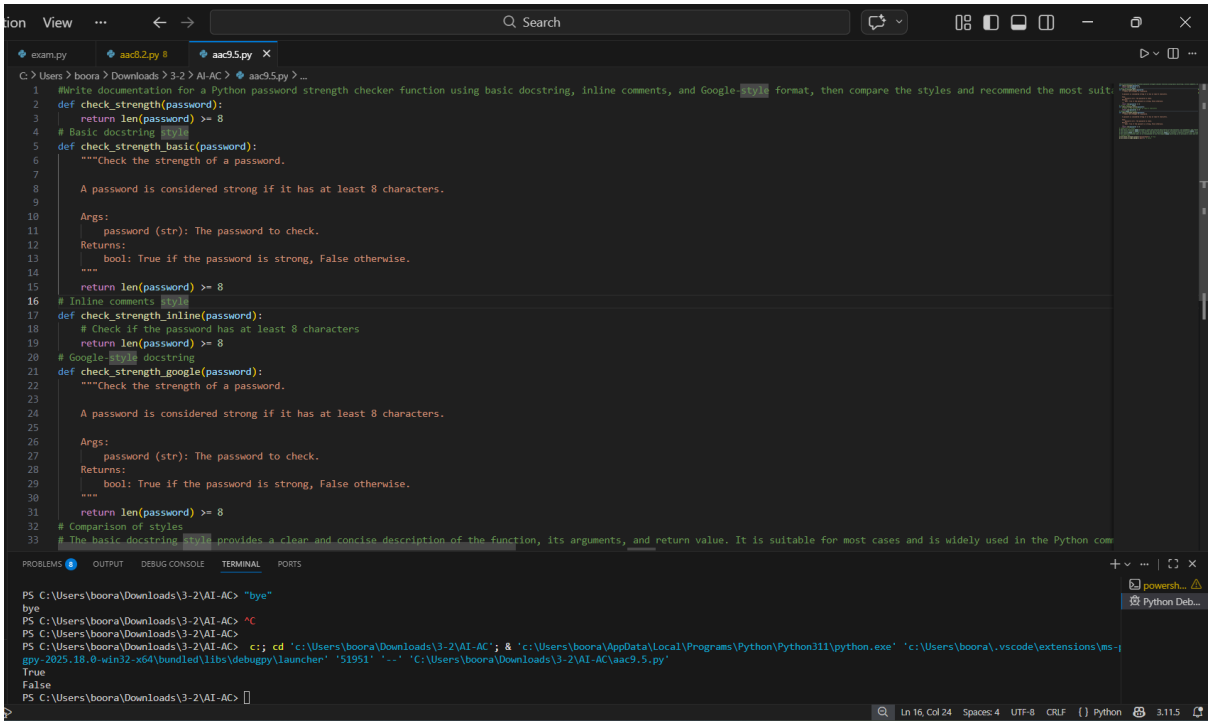
Google style.

2. Compare documentation styles for security-related code.

3. Recommend the most appropriate style.

Prompt:

#Write documentation for a Python password strength checker function using basic docstring, inline comments, and Google-style format, then compare the styles and recommend the most suitable one for security-related code.



Problem 3: Math Utilities Module

Task:

1. Create a module math_utils.py with functions:

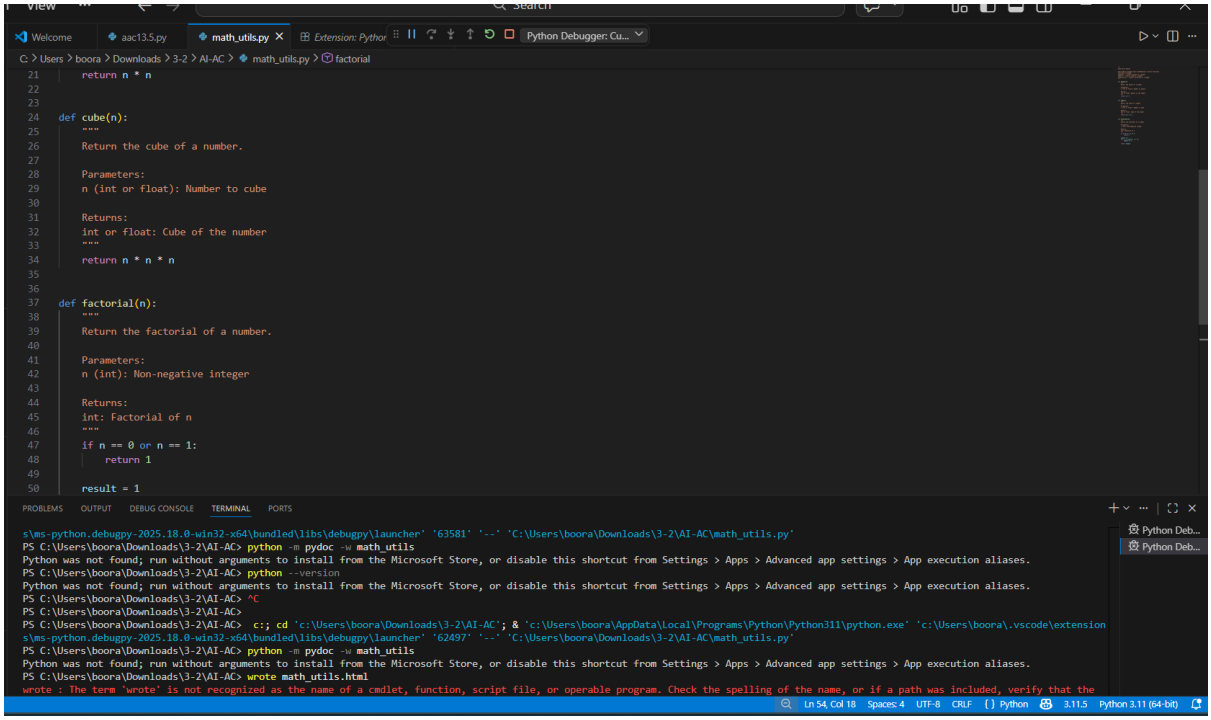
o square(n)

o cube(n)

o factorial(n)

2. Generate docstrings automatically using AI tools.

3. Export documentation as an HTML file.



Problem 4: Attendance Management Module

Task:

1. Create a module attendance.py with functions:

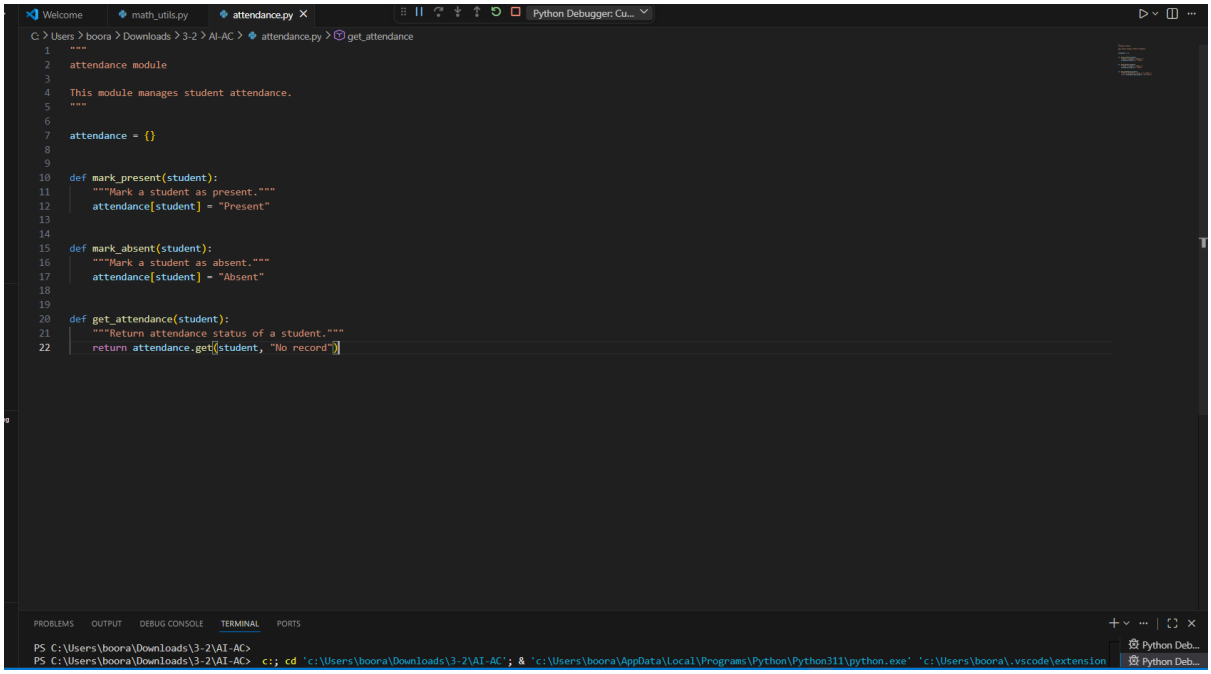
o mark_present(student)

o mark_absent(student)

o get_attendance(student)

2. Add proper docstrings.

3. Generate and view documentation in terminal and browse


Welcome
math_utils.py
attendance.py
Python Debugger: Cu...
Users > boora > Downloads > 3-2 > AI-AC > attendance.py > get_attendance
"""
attendance module
This module manages student attendance.
"""
attendance = {}
def mark_present(student):
"""Mark a student as present."""
attendance[student] = "Present"
def mark_absent(student):
"""Mark a student as absent."""
attendance[student] = "Absent"
def get_attendance(student):
"""Return attendance status of a student."""
return attendance.get(student, "No record")
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
PS C:\Users\boora\Downloads\3-2\AI-AC>
PS C:\Users\boora\Downloads\3-2\AI-AC> c:; cd 'c:\Users\boora\Downloads\3-2\AI-AC'; & 'c:\Users\boora\AppData\Local\Programs\Python\Python311\python.exe' 'c:\Users\boora\.vscode\extension
Python Deb...
Python Deb...